Department of Software Engineering

Quiz 1

Course Code: SE214

Course Title: Algorithms Design & Analysis

Time: 45 minutes **Total Marks: 15**

Name:

Student ID: Section:

Date: Semester:

Marks: Question 1: 3 + Merge Sort: 5 + Insertion Sort: 5 + Attendance: 2.

Instructions: Answer all questions in the blank spaces provided. Show sorting steps and briefly justify complexity results.

Question 1 Algorithm and complexity analysis **3 marks**

(a) Define an algorithm. **[1]**

(b) Determine the time complexity and space complexity of the following nested loop in terms of n. Assume n is a positive integer and each arithmetic operation takes constant time. **[1 + 1]**

```
int count = 0;
for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        count++;
    }
}
```

Question 2 Merge Sort and complexity analysis **5 marks**

(a) Sort **[38, 27, 43, 3, 9, 10]** in ascending order using Merge Sort. Show the splitting and merging steps. **[3]**

(b) State the best, average, and worst-case time complexities of Merge Sort. Briefly justify your answer. **[2]**

Question 3 Insertion Sort and complexity analysis **5 marks**

(a) Sort **[9, 5, 1, 4, 3]** in ascending order using Insertion Sort. Show the array after each pass. **[3]**

(b) State the best, average, and worst-case time complexities of Insertion Sort. Explain which input order produces the best and worst cases. **[2]**

For instructor use only

Q1: / 3 Q2: / 5 Q3: / 5 Attendance: / 2

Total: / 15 Instructor Signature: